Expected Performance Index Report

Methodology and Purpose

To decouple tree age from environmental suitability, we employ a Gamma Generalised Additive Model (GAM). Unlike standard regression, a Gamma GAM respects the biological reality that tree size must be positive and that variance increases as trees mature (heteroscedasticity).

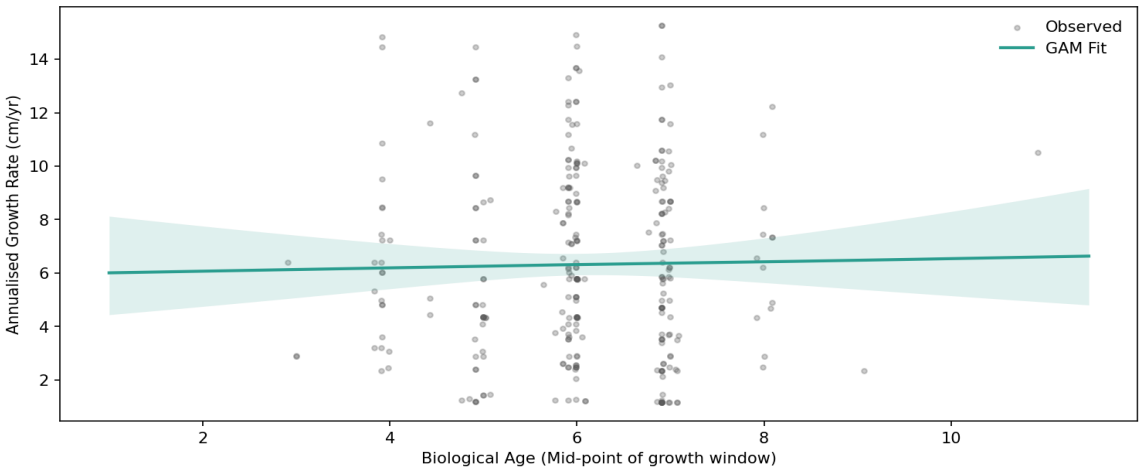
By using a Log-Link function, we capture the natural growth curve. The resulting Expected Performance Index (EPI) is the ratio of actual growth to the model's predicted baseline. This creates a standardized metric: an EPI of 1.2 indicates growth 20% above the species average for that age, isolating the environmental signal for use in our MCDA weight extraction.

Model Fitting Summary

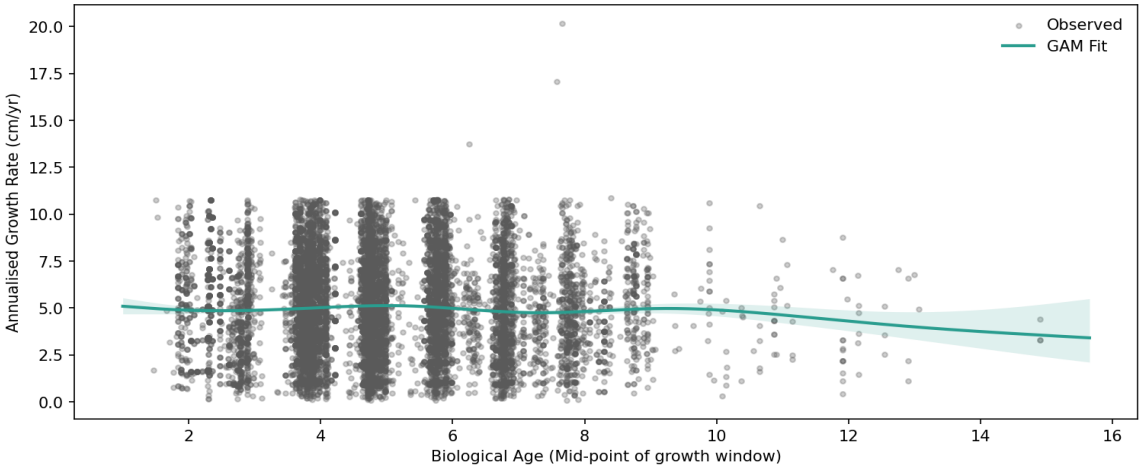
Species	Points	AIC	Deviance	EDOF
Ai-bubur (mutin)	301	1612.34	206.78	2.05
Ai-kakeu	14004	67436.10	9917.42	6.31
Ai-kamelli	573	2255.53	465.05	5.50
Ai-mahony	189684	887181.44	130631.42	9.74
Ai-na	307	1240.52	259.40	4.33
Ai-nitas	947	3200.89	745.50	2.11
Ai-ru	4558	19336.48	3403.65	5.27
Ai-saria	898	3750.53	581.07	9.74
Ai-teka	19079	84238.02	15244.86	5.58

Species Specific Fits

Species: Ai-bubur (mutin)

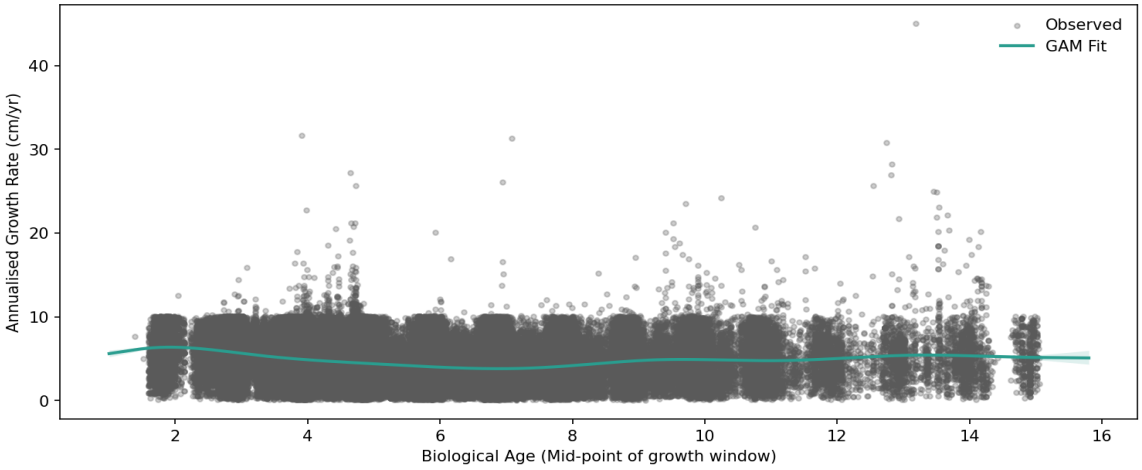


Species: Ai-kakeu

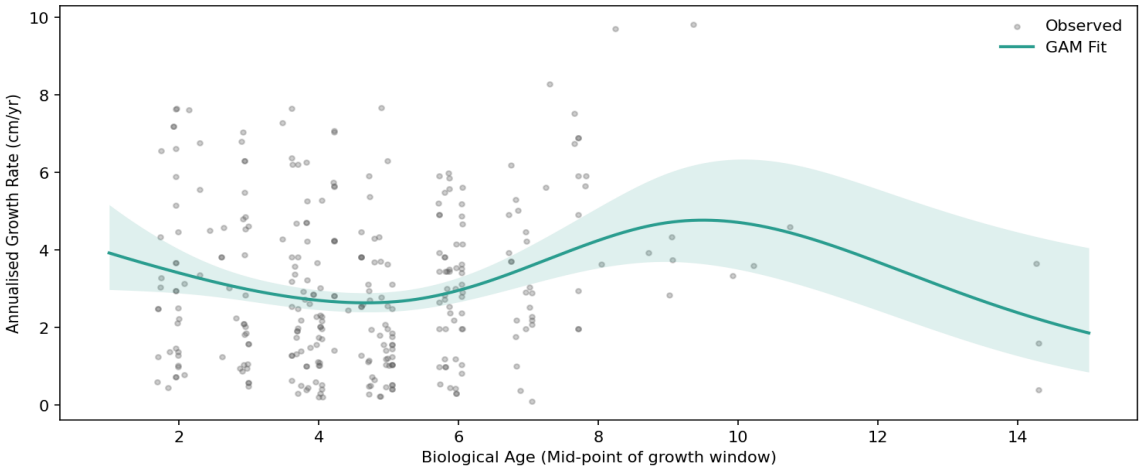


Species: Ai-kamelli

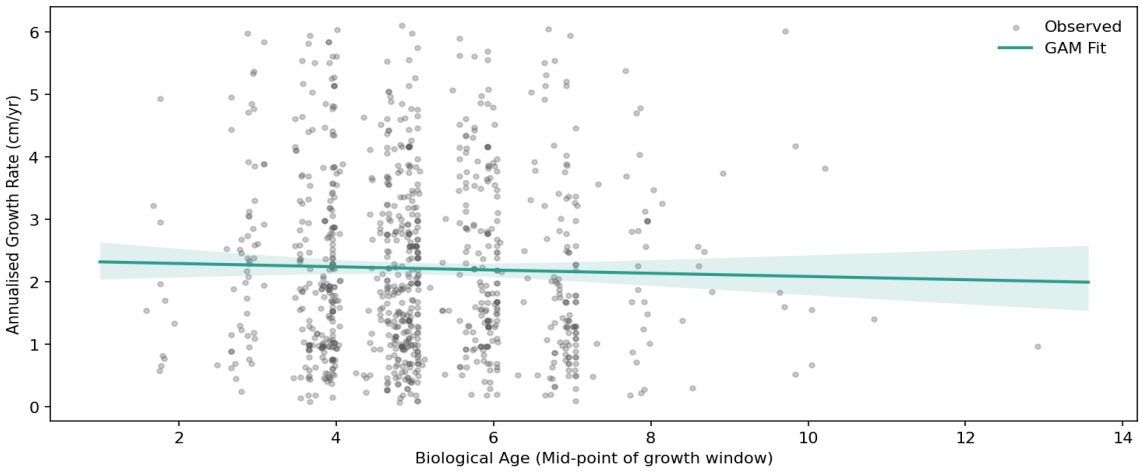
Species: Ai-mahony



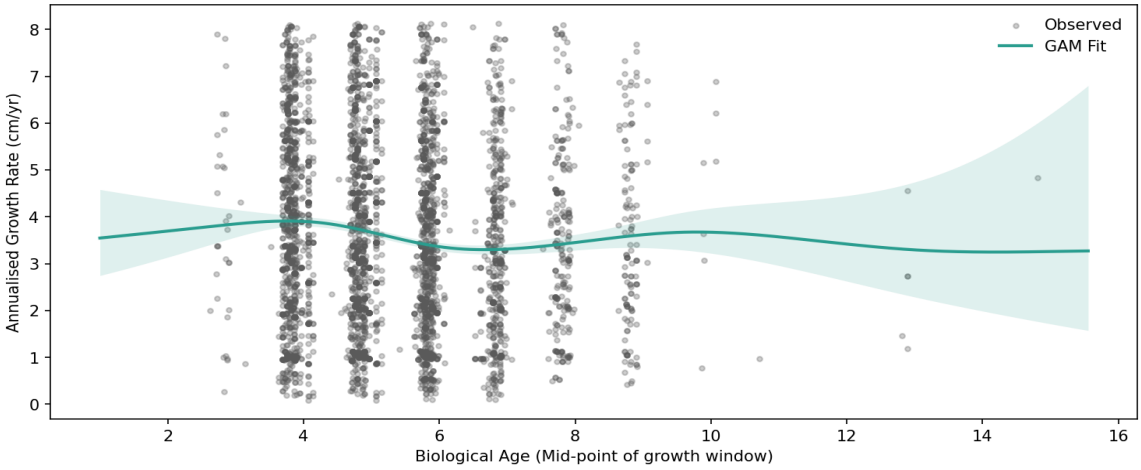
Species: Ai-na



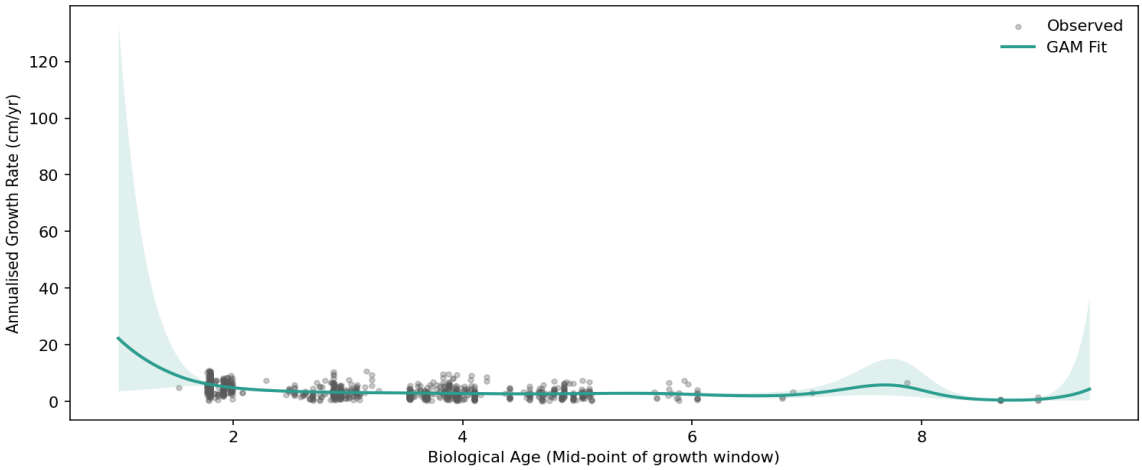
Species: Ai-nitas



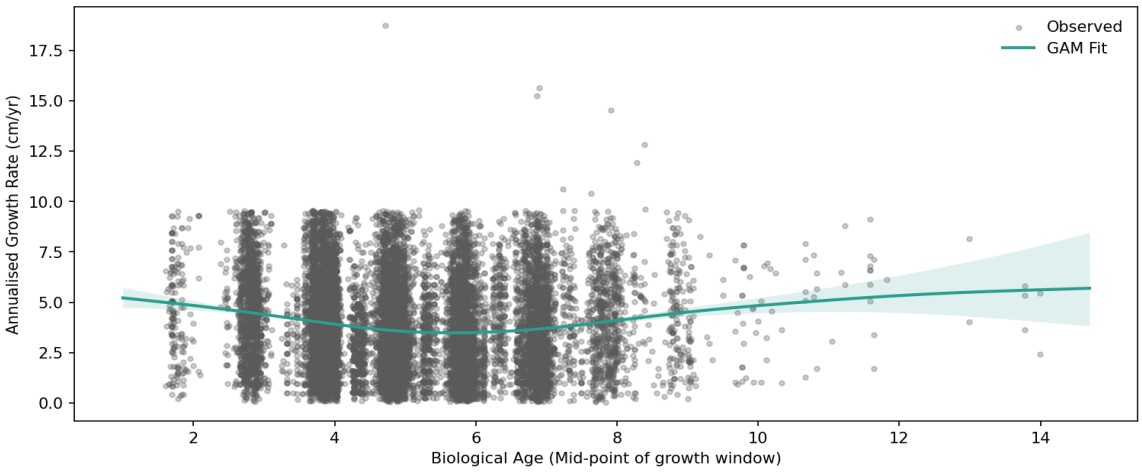
Species: Ai-ru



Species: Ai-saria

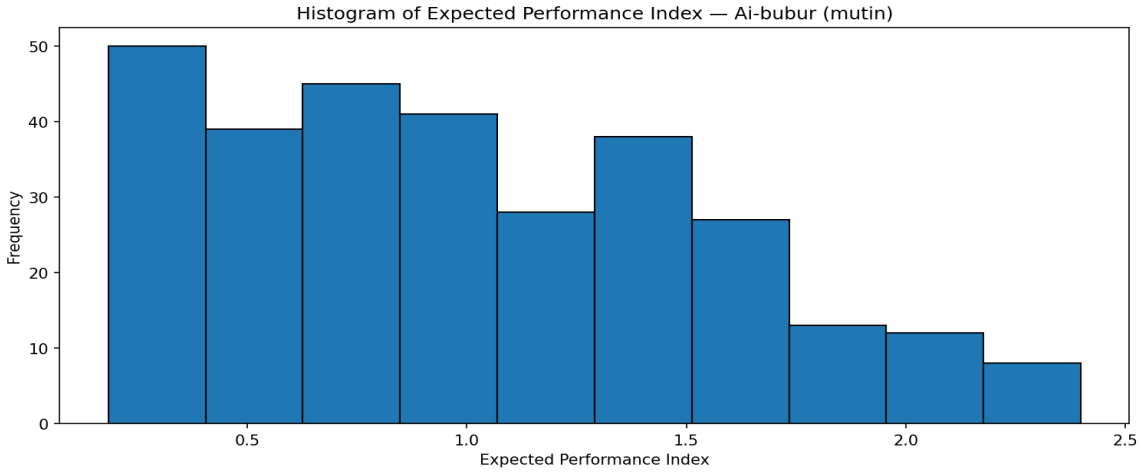


Species: Ai-teka

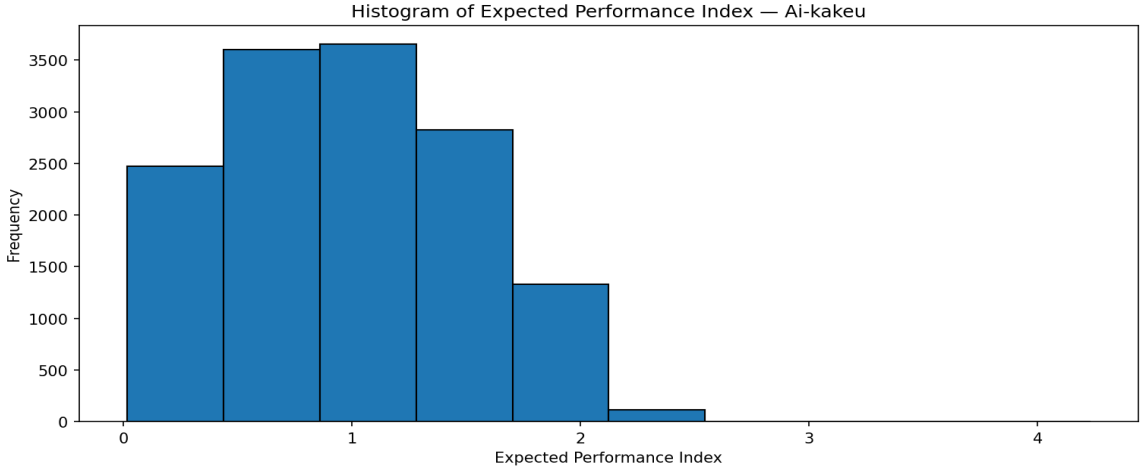


Species Specific Expected Performance Index Histograms

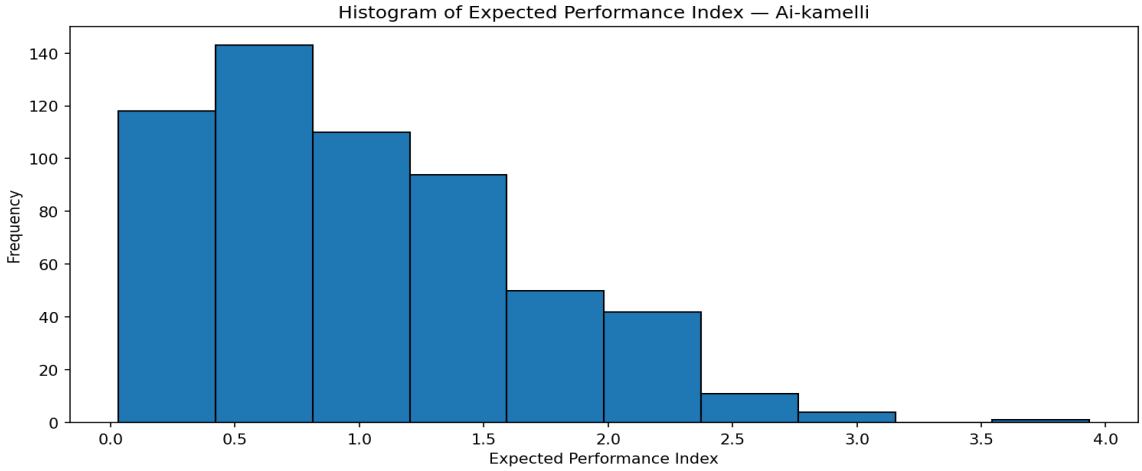
Species: Ai-bubur (mutin)



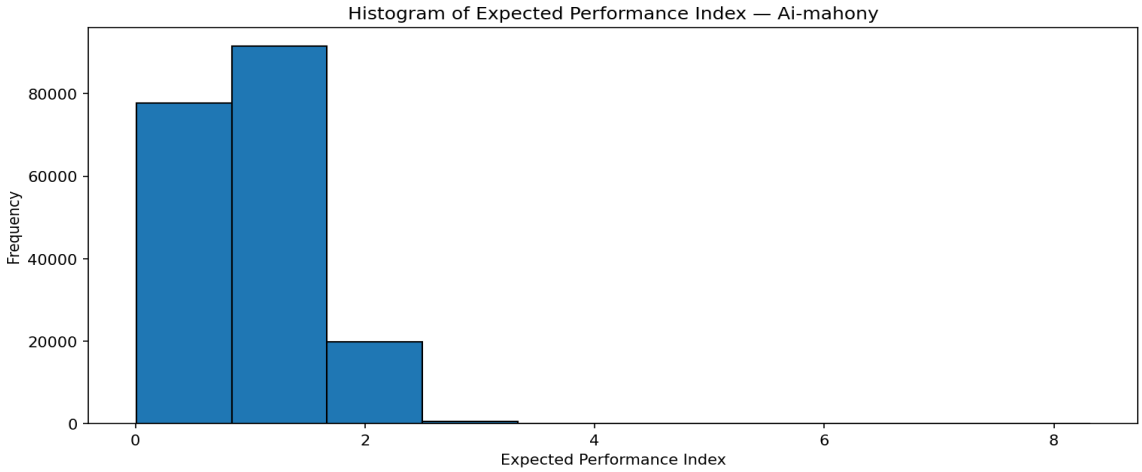
Species: Ai-kakeu



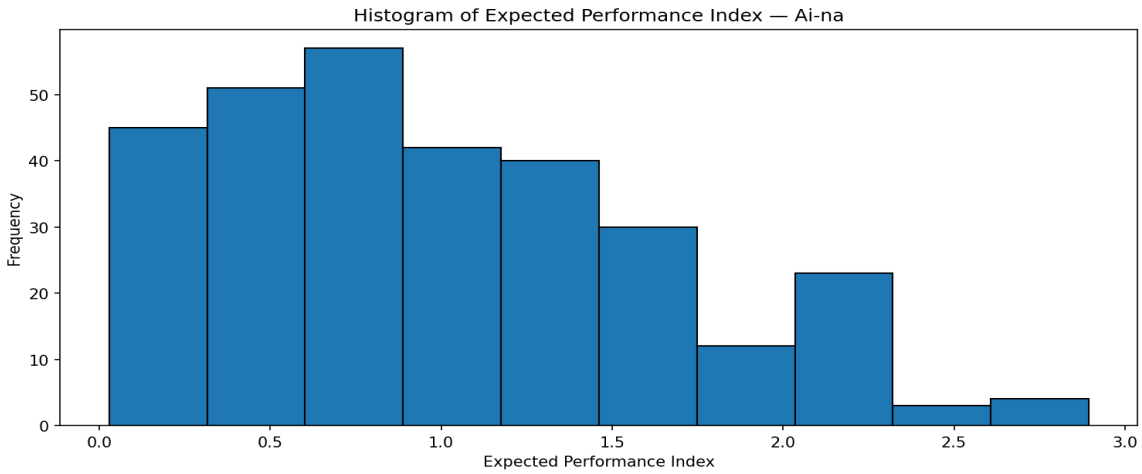
Species: Ai-kamelli



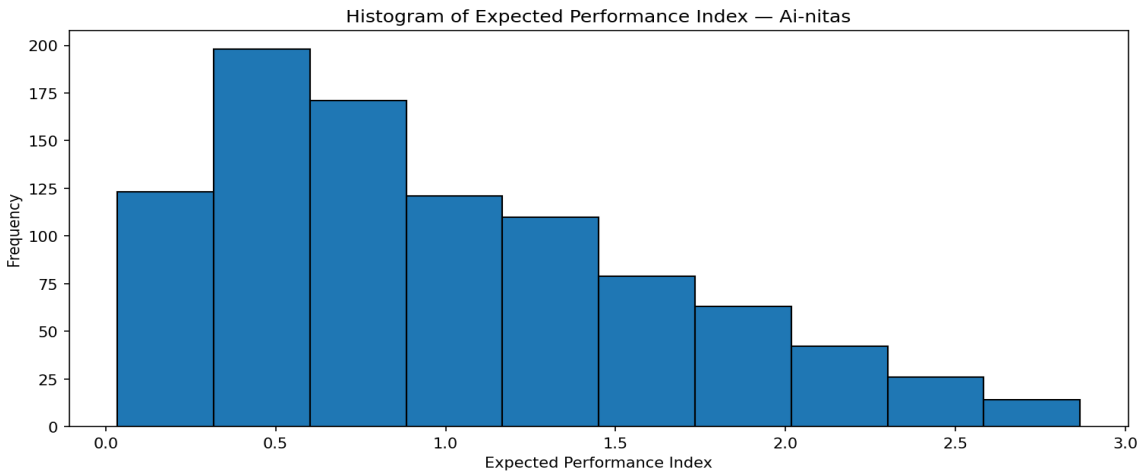
Species: Ai-mahony



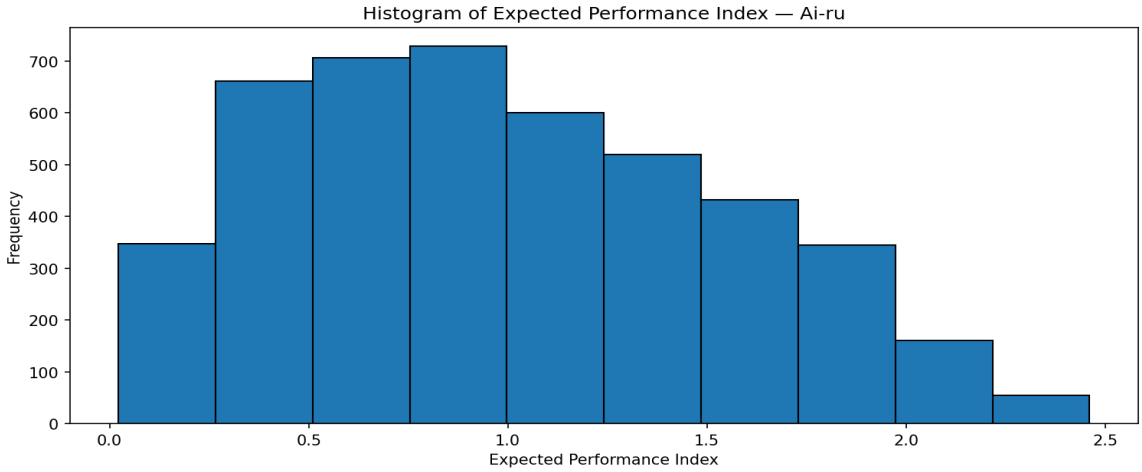
Species: Ai-na



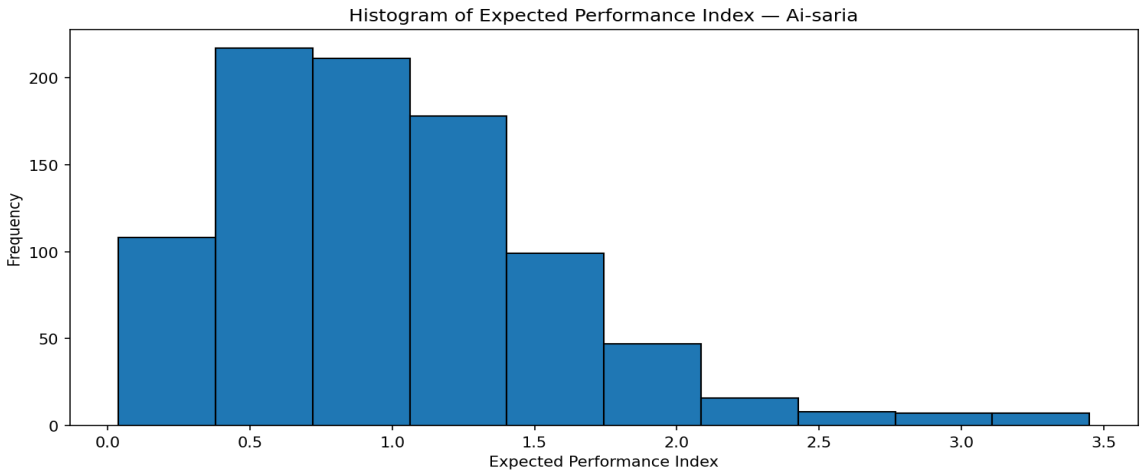
Species: Ai-nitas



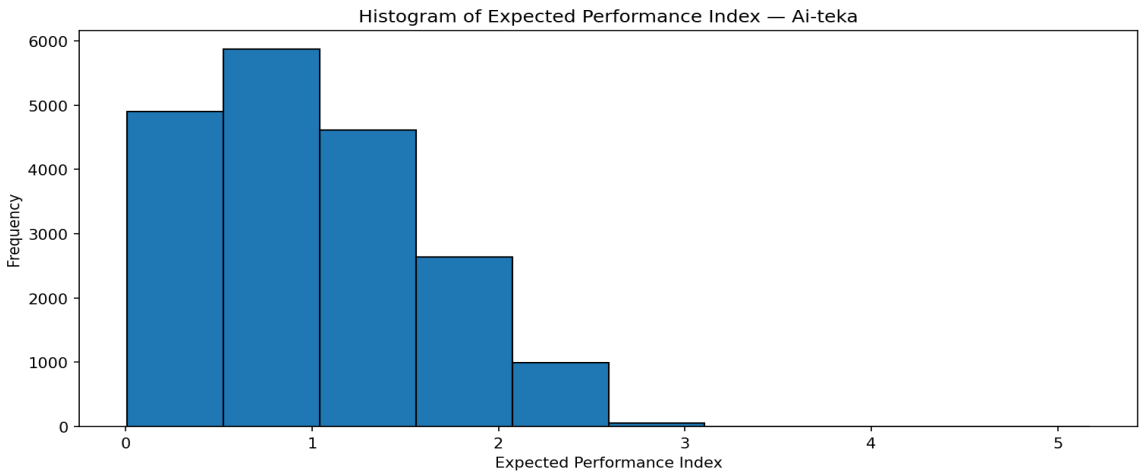
Species: Ai-ru



Species: Ai-saria



Species: Ai-teka



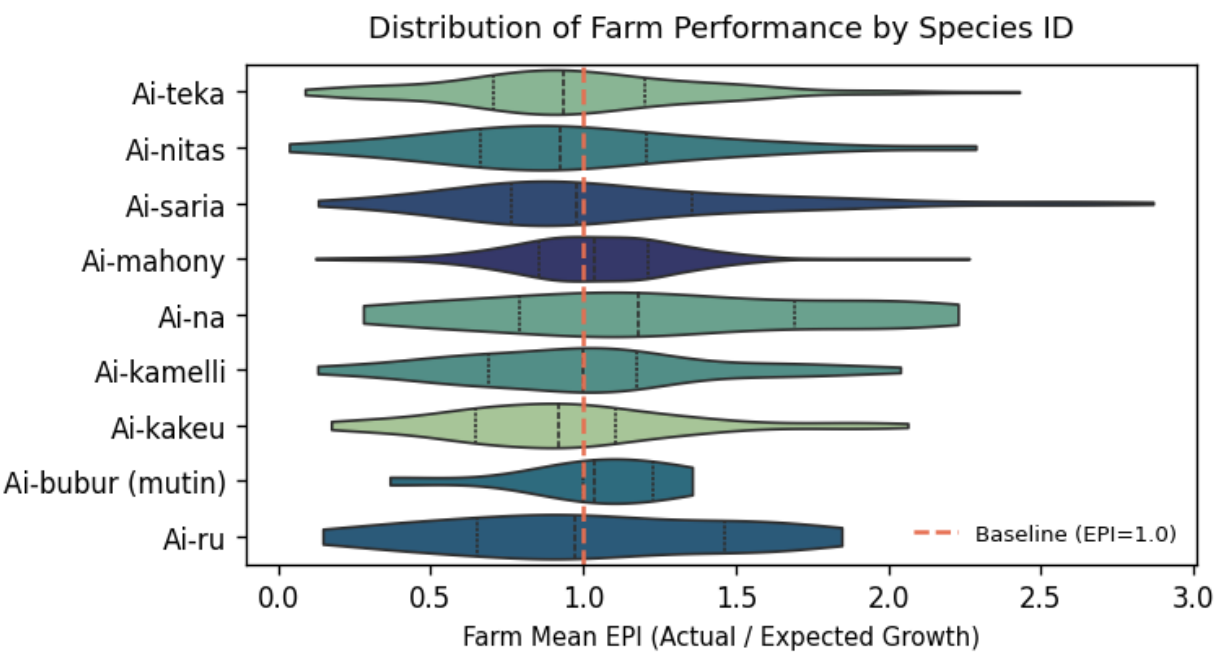
Farm-Level Performance Aggregation

Aggregate to the Farm Level

Before feeding the Expected Performance Index (EPI) into the final weight-extraction GAM, the tree-level data is grouped by farm identifier (e.g., farm_id) and species_id. This achieves two vital things:

1. Noise Reduction: Individual trees exhibit random variance (e.g., one tree got struck by lightning, another had an isolated fungal infection). Averaging the EPI of all trees of a specific species on a single farm smooths out that individual noise, resulting in a highly stable "Farm Performance Score."

2. Alignment with Features: The environmental features (Rainfall, Temperature, Soil suitability) are scored at the farm level, not the individual tree level. The target variable (EPI) must perfectly align with these inputs for accurate mathematical extraction.

Interpreting the Violin Plot

The plot below illustrates the distribution of these aggregated Farm EPI scores across different species. It highlights the density of farm performance: wider sections indicate a high concentration of farms achieving that specific score. The dashed vertical line at 1.0 represents the biological baseline (where actual growth perfectly matches expected growth).